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1 author:

Pieter Chys
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Application of geometric algebra for the description of polymer conformations

Pieter Chys[a)]
Department of Chemistry, Katholieke Universiteit Leuven, Division Biochemistry, Molecular and Structural Biology, Group Protein Chemistry, Celestijnenlaan 200 G, 3001 Leuven-Heverlee, P.O. Box 2413, Belgium

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In this paper a Clifford algebra-based method is applied to calculate polymer chain conformations. The approach enables the calculation of the position of an atom in space with the knowledge of the bond length (l), valence angle (θ), and rotation angle (φ) of each of the preceding bonds in the chain. Hence, the set of geometrical parameters $\{l_i, \theta_i, \varphi_i\}$ yields all the position coordinates $\mathbf{p}_i$ of the main chain atoms. Moreover, the method allows the calculation of side chain conformations and the computation of rotations of chain segments. With these features it is, in principle, possible to generate conformations of any type of chemical structure. This method is proposed as an alternative for the classical approach by matrix algebra. It is more straightforward and its final symbolic representation considerably simpler than that of matrix algebra. Approaches for realistic modeling by means of incorporation of energetic considerations can be combined with it. This article, however, is entirely focused at showing the suitable mathematical framework on which further developments and applications can be built. © *2008 American Institute of Physics*.
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I. INTRODUCTION

At present the field of polymer physics has evolved into a state of art where a sound theoretical basis is present. Major steps in theoretics were made by scientists as Flory[1] and de Gennes[2] who propelled the field. A good general account of the present state is given in Ref. 3.

Here we apply the geometric algebra as defined by Hestenes and co-workers[4–6] to generate polymer conformations. This paper thus focuses on a new method to calculate conformations in the context of polymer modeling. The main principle of this geometric algebra approach is to define multivector objects which can then be transformed according to local rotations along the chain. The result is the possiblity to efficiently generate chain segment rotations in a recursive sense.

More in detail, the method presented here allows us to calculate position coordinates on the basis of the following geometrical parameters along an entire polymer chain: Bond length l_i, valence angle θ_i, and dihedral angle φ_i where the index i refers to each of the constituting elements. One can generate by use of this method any type of chemical structure. It is particularly useful that these parameters appear in the relevant equations in a transparent way. In geometric algebra, e.g., the rotation through a dihedral angle φ_i along an axis is represented very naturally in the equation of a spinor. No such similar relation is present in matrix algebra and consequently the method can be used as an alternative for the existing matrix methods which remain very powerful tools for modeling polymer structures.[7] A good account of the application of these matrix methods for polymer physics is given by Flory's standard work.[7] Later in the paper (Sec. III E) we will make a concise comparison between matrix algebra as presented in Flory's text and geometric algebra to illustrate the differences.

Since any type of molecule or macromolecule can be computed, each scientific field in which conformations have to be calculated can potentially benefit from this method. Two fields deserve special attention. In the first place it is of major importance for biologists and biochemists who are active in biological modeling,[8,9] both academically and industrially. Protein and ligand structures can be easily generated. Furthermore the possibility to rotate chain segments allows conformational manipulation which is important for optimalization of the conformation of a given chemical structure. The second field is the nonbiological macromolecular field. A large fraction of the chemical industry covers the subfield of polymer physics and at universities a considerable amount of related research work is done. Thus the method will be of interest for chemical engineers and polymer scientists who are concerned with conformational details of polymers. Apart from those two major fields the method can be applied by anyone involved in research of chemical structures.

The practical implementation should be twofold: (i) The incorporation of algorithms—following the presented method—in modeling programs would be beneficial; this implementation should be at the basic level since it concerns the computational scheme to be used. (ii) Small-scale use is also possible; e.g., we wrote a small program (see appendixes) which performs computations along the theoretical outline presented here. This type of small-scale use is more flexible and will allow the user to put more accents in his

[a)]Telephone: +32-16-327696. FAX: +32-16-327978. Electronic mail: pieter.chys@chem.kuleuven.be.

work. It also allows combination with other methods (e.g., energetic approaches) which might not be supported by standard modeling software. Some mathematical packages (e.g., GABLE) are also available which allow such calculations.

It is illustrative for geometric algebra that it is entering fields such as those of computer vision and robotics because of a higher computing efficiency.[10,11] The NASA has also supported basic research of possible applications of geometric algebra for space programs.[12,13] Furthermore geometric algebra has found applications in quantum mechanics and relativistic theory (a straightforward generalization) because of its more concise form.[14,15] Hence, we believe that introduction of this algebra is of major importance for the fields of polymer physics and biochemistry.

The outline of the article is as follows. In Sec. II a brief summary is given of the most important elements of geometric algebra which are necessary to understand the derivation and use of main formulas. Section III is the core of the paper. In Sec. III A conventions on correct referencing to chain elements are given and Sec. III B derives the main formulas to generate atom positions. Section III C is an annex to Sec. III B which extends the formulas for side chains. In Sec. III D an equation to describe rotation of large chain segments is derived. Sec. III E compares geometric algebra with matrix algebra in general. Their computational efficiency is also studied in more detail in this section. In Sec. IV a calculation is made to illustrate the approach. Finally, in Sec. V the conclusions are drawn. The appendixes, contain supplementary material. It deals with details concerning the calculations and discusses the MATLAB program we wrote for calculations.

II. CONCEPTS OF GEOMETRIC ALGEBRA

In the geometric algebra the noncommutative geometric product is of major importance. We will use the outer product (denoted with $\wedge$) because of its geometrical meaning (a bivector representing a plane) which can then be calculated by the geometric product. In addition we will take advantage of spinors since they express rotations in a natural way. Spinors are embedded in the Clifford algebra as the even subalgebra (also called spinor algebra). This is an illustration of the fact that geometric algebra unifies other separate algebras and as a consequence their strengths. The mathematical details of the algebra are out of the scope of this paper but can be found in the reference works.[4–6] We will follow the notation as given in Ref. 4 and this includes that multivectors will be presented in boldface-type symbols. Two exceptions are made: We substitute the unit trivector i by ι to avoid confusion with the extensive use of the subscript i and present spinors as boldface-type symbols.

The geometric product of two 1-vectors **a** and **b** is **ab**; this product is noncommutative,

$$\mathbf{ab} \neq \mathbf{ba}.$$

A 1-vector is mathematically the same as a classical vector in Gibbs vector algebra.[16] This also implies the similarity in physical meaning. However, whereas vectors form the basis of Gibbs vector algebra, 1-vectors are only a special case of

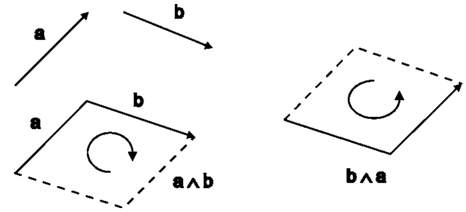

FIG. 1. Illustration of 1-vectors (**a** and **b**) and 2-vectors ($\mathbf{a}\wedge\mathbf{b}$ and $\mathbf{b}\wedge\mathbf{a}$) as used in geometric algebra. The two bivectors have a different orientation: $\mathbf{a}\wedge\mathbf{b}$ has clockwise while $\mathbf{b}\wedge\mathbf{a}$ has counterclockwise orientation (viewed from above).

multivectors in geometric algebra. For instance, a 2-vector or bivector in geometric algebra represents physically an orientated plane which is defined by two 1-vectors (Fig. 1). More generally geometric algebra thus incorporates the concept of multivectors. In physical applications multivectors up to a degree of 4 (4-vectors) are common (e.g., 4-vectors are used to describe relativity theory). For this text the use of 2-vectors is necessary due to their physical meaning whereas the unit 3-vector (trivector) is used in a manipulative and computational way for obtaining appropriate equations. We now consider in more detail the geometric product **ab** of two 1-vectors which is given by

$$\mathbf{ab} = \mathbf{a}\cdot\mathbf{b} + \mathbf{a}\wedge\mathbf{b}. \tag{1}$$

In this equation $\mathbf{a}\cdot\mathbf{b}$ is the interior (inner) product and $\mathbf{a}\wedge\mathbf{b}$ the exterior (outer) product. The interior product has the same meaning as in Gibbs vector algebra (scalar product) and is in addition a degree lowering operation: The result of $\mathbf{a}\cdot\mathbf{b}$ is a scalar (0-vector). The outer product, on the contrary, is degree raising and thus $\mathbf{a}\wedge\mathbf{b}$ is a 2-vector. It represents the directional plane formed by the 1-vectors **a** and **b**. The use of Eq. (1) on applying to **ab** and **ba** yields the following important relations:

$$\begin{aligned}\mathbf{a}\cdot\mathbf{b} &= \tfrac{1}{2}(\mathbf{ab}+\mathbf{ba}),\\ \mathbf{a}\wedge\mathbf{b} &= \tfrac{1}{2}(\mathbf{ab}-\mathbf{ba}).\end{aligned} \tag{2}$$

In this way, both products can be calculated by means of the geometric product. As in Gibbs vector algebra, multivectors (and thus 1-, 2-, and 3-vectors) can be written in terms of their corresponding unit multivectors. If we use the unit 1-vectors $\boldsymbol{\sigma_1}$, $\boldsymbol{\sigma_2}$, and $\boldsymbol{\sigma_3}$ as basis $\{\boldsymbol{\sigma_1}, \boldsymbol{\sigma_2}, \text{and } \boldsymbol{\sigma_3}\}$ (similar to $\bar{i}$, $\bar{j}$, and $\bar{k}$ in Gibbs algebra) the 1-vector **a** can be expressed as

$$\mathbf{a} = a_1\boldsymbol{\sigma_1} + a_2\boldsymbol{\sigma_2} + a_3\boldsymbol{\sigma_3},$$

in which a_1, a_2, and a_3 are scalars. By an appropriate choice of the unit 1-vectors (orthonormal relative orientation) $\mathbf{a}\cdot\mathbf{b}$ becomes 0 and by use of Eq. (1) we get

$$\begin{aligned}\boldsymbol{\sigma_1\sigma_2} &= \boldsymbol{\sigma_1}\wedge\boldsymbol{\sigma_2} \equiv \mathbf{i}_3,\\ \boldsymbol{\sigma_1\sigma_3} &= \boldsymbol{\sigma_1}\wedge\boldsymbol{\sigma_3} \equiv \mathbf{i}_2,\\ \boldsymbol{\sigma_2\sigma_3} &= \boldsymbol{\sigma_2}\wedge\boldsymbol{\sigma_3} \equiv \mathbf{i}_1,\end{aligned} \tag{3}$$

in which $\mathbf{i}_1$, $\mathbf{i}_2$, and $\mathbf{i}_3$ are the three unit bivectors which one can imagine as laying in the YZ-, XZ-, and XY-plane of a corresponding orthonormal Cartesian system. Any arbitrary bivector $\mathbf{A}$ can also be written as a linear combination of those three unit bivectors as follows:

$$\mathbf{A} = A_1\mathbf{i}_1 + A_2\mathbf{i}_2 + A_3\mathbf{i}_3.$$

As the previous equations show, the interior product has the same meaning as in Gibbs vector algebra (e.g., $\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 = 0$). When taking the geometric product of $\boldsymbol{\sigma}_1$ with itself, the bivector $\boldsymbol{\sigma}_1 \wedge \boldsymbol{\sigma}_1$ equals 0—this does not represent a plane—and Eq. (1) becomes

$$\boldsymbol{\sigma}_1\boldsymbol{\sigma}_1 = \boldsymbol{\sigma}_1^2 = \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_1 = 1,$$

in which the interior product is maximal ($\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_1 = |\boldsymbol{\sigma}_1||\boldsymbol{\sigma}_1|$). If we extend the concepts and equations to trivectors, the unit 3-vector is equal to

$$\iota = \boldsymbol{\sigma}_1 \wedge \boldsymbol{\sigma}_2 \wedge \boldsymbol{\sigma}_3 = \boldsymbol{\sigma}_1\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3. \tag{4}$$

Furthermore each trivector can be written in its unit multivector components or thus

$$\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} = \lambda\iota,$$

in which λ is a scalar.

The product of ι with a multivector $\mathbf{M}$ gives its dual $\mathbf{M}'$ (also a multivector) or

$$\mathbf{M}' = \iota\mathbf{M}.$$

For a 1-vector $\mathbf{a}$ the dual $\mathbf{M}'$ is a bivector $\mathbf{A}$,

$$\mathbf{A} = \iota\mathbf{a}.$$

A useful relation between Gibbs vector algebra and the outer product of the geometric algebra is given by

$$\mathbf{a} \times \mathbf{b} = -\iota\mathbf{a} \wedge \mathbf{b} \tag{5}$$

or, otherwise stated $\mathbf{a} \times \mathbf{b}$ is the negative of the dual of the bivector $\mathbf{a} \wedge \mathbf{b}$. Further manipulation of Eq. (5) yields

$$\mathbf{a} \times \mathbf{b} = -\iota\mathbf{a} \wedge \mathbf{b},$$

$$\iota(\mathbf{a} \times \mathbf{b}) = -\iota^2\mathbf{a} \wedge \mathbf{b},$$

$$\begin{aligned}\boldsymbol{\sigma}_1\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3(\mathbf{a} \times \mathbf{b}) &= -(\boldsymbol{\sigma}_1\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3\boldsymbol{\sigma}_1\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3)\mathbf{a} \wedge \mathbf{b}\\ &= +(\boldsymbol{\sigma}_2\boldsymbol{\sigma}_1\boldsymbol{\sigma}_3\boldsymbol{\sigma}_1\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3)\mathbf{a} \wedge \mathbf{b}\\ &= -(\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3\boldsymbol{\sigma}_1^2\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3)\mathbf{a} \wedge \mathbf{b}\\ &= -(\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3\boldsymbol{\sigma}_2\boldsymbol{\sigma}_3)\mathbf{a} \wedge \mathbf{b}\\ &= (\boldsymbol{\sigma}_2^2\boldsymbol{\sigma}_3^2)\mathbf{a} \wedge \mathbf{b},\end{aligned}$$

$$\iota(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \wedge \mathbf{b}.$$

It follows that $\mathbf{a} \wedge \mathbf{b}$ is the dual of $\mathbf{a} \times \mathbf{b}$ and since $\mathbf{a} \wedge \mathbf{b}$ is a bivector, the vector product $\mathbf{a} \times \mathbf{b}$ is a 1-vector which we know to be true from Gibbs vector algebra. Note also that we have proven by means of this example that if $\mathbf{A} = \iota\mathbf{a}$ is valid, also $\mathbf{a} = -\iota\mathbf{A}$ is valid.

To this point we only considered multivectors which could be expressed in terms of their unit multivectors. In its most general form a multivector $\mathbf{M}$ (e.g., n-vector) can, however, be written as

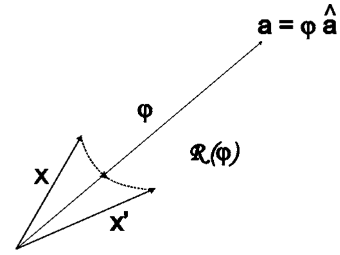

FIG. 2. Schematic representation of a spinor operator $\mathcal{R}$. The spinor operator $\mathcal{R}(\varphi)$ performs a rotation of $\mathbf{x}$ through an angle φ and results in the 1-vector $\mathbf{x}'$. A positive value of φ corresponds to a clockwise rotation if we look along the axis of the 1-vector $\mathbf{a}$ towards its endpoint (right hand rule). The value of φ in this figure would be +180°.

$$\mathbf{M} = \sum_{i=0}^{n} \langle \mathbf{M} \rangle_i, \tag{6}$$

and thus, if $n=3$, $\mathbf{M}$ becomes

$$\mathbf{M} = \langle \mathbf{M} \rangle_0 + \langle \mathbf{M} \rangle_1 + \langle \mathbf{M} \rangle_2 + \langle \mathbf{M} \rangle_3,$$

in which the subscript i refers to each i-vector component.

With the help of Eq. (6) we now are able to describe spinors. The bivector $\mathbf{S}$ of the form

$$\mathbf{S} = \alpha + \iota\mathbf{b},$$

in which α is a scalar ($\langle \mathbf{M} \rangle_0$), the 1-vector part ($\langle \mathbf{M} \rangle_1$) is lacking, and the alternative notation of $\iota\mathbf{b}$ for its 2-vector component ($\langle \mathbf{M} \rangle_2$) is used, is a spinor. The 2-vectors which obey this general equation (spinors) describe efficiently rotations in space. The spinor $\mathbf{R}$ is equal to

$$\mathbf{R} = e^{(1/2)\mathbf{A}} = e^{(1/2)\iota\mathbf{a}},$$

in which $\mathbf{A}$ is a bivector and the dual of 1-vector $\mathbf{a}$. Physically $\mathbf{a}$ represents the rotation axis about which the spinor $\mathbf{R}$ will perform a rotation of a geometric object (e.g., a 1-vector). Further elaboration yields

$$\mathbf{R} = e^{(1/2)\iota\varphi\hat{\mathbf{a}}},$$

in which $\hat{\mathbf{a}}$ is the unit 1-vector in the direction of $\mathbf{a}$ and φ the magnitude of $\mathbf{a}$ or $\varphi = |\mathbf{a}|$. The value of φ is equal to the rotation angle through which a geometric object will be rotated by the spinor $\mathbf{R}$. For a positive value of φ the rotation sense is clockwise following the convention given in Fig. 2. The right hand rule applies. In nonexponential form $\mathbf{R}$ is

$$\mathbf{R} = \cos\left(\frac{\varphi}{2}\right) + \iota\hat{\mathbf{a}}\sin\left(\frac{\varphi}{2}\right).$$

The concept of a spinor comes into a useful form by means of the following equation:

$$\mathbf{x}' = \mathbf{R}^\dagger\mathbf{x}\mathbf{R} = \mathcal{R}\mathbf{x}, \tag{7}$$

where the reverse spinor $\mathbf{R}^\dagger$ is only different from $\mathbf{R}$ by the sign in the exponent

$$\mathbf{R}^\dagger = e^{-(1/2)\mathbf{A}}.$$

For convenience we have introduced in Eq. (7) $\mathcal{R}\mathbf{x}$ as notation for $\mathbf{R}^\dagger\mathbf{x}\mathbf{R}$ which is also satisfactory from the point of view that $\mathcal{R}$ is a linear operator. Thus the operation of spinor

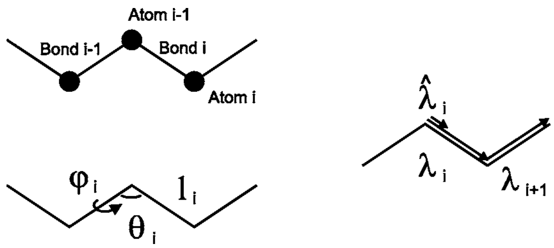

FIG. 3. Conventions on the numbering of atoms, bonds, and angles in the text. An important element is that the dihedral angle φ_i refers to a rotation around the previous bond and not the actual bond of the atom. The involved valence angle is also not located at the atom itself but at the preceding atom. The λ_i associated with bond i (upper left) is depicted on the right in which the unit 1-vector $\hat{\lambda}_i$ is also represented.

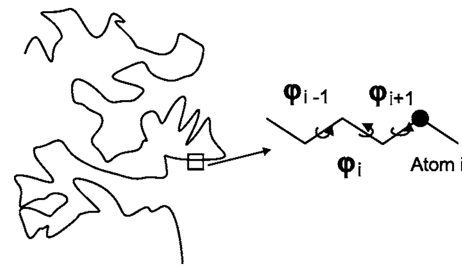

FIG. 4. A polyethylene chain and the notation involved for the rotation angles.

operator $\mathcal{R}$ on 1-vector $\mathbf{x}$ leads to the 1-vector $\mathbf{x}'$ which results from rotation of $\mathbf{x}$ through an angle φ along the axis of $\mathbf{a}$ (Fig. 2).

It should be mentioned that spinors were discovered by Hamilton in the form of quaternions and their calculation properties (1843). This was while he was searching for an extension of complex numbers to three dimensions. The quaternionic algebra can now be derived as a subalgebra of geometric algebra (Clifford algebra)[4] and actually we are using quaternions but derived in the framework of geometric algebra. This fact illustrates the encompassing nature of geometric algebra.

III. APPLICATION OF THE ALGEBRA

A. The setup of a chain

Given a polymer chain, we start numbering the main chain atoms from one of its endpoints (1,2,…). The ith atom is now referred to as atom i. The bond between atoms $i-1$ and i is bond i. Valence angle θ_i is located at atom $i-1$ and rotation angle φ_i describes the rotation around the bond joining atoms $i-2$ and $i-1$. Hence, φ_i is associated with bond $i-1$ and θ_i with atom $i-1$ (Fig. 3).

We chose this notation because changes in values of φ_i and θ_i affect the position of atom i. The choice is thus functional and not physically based.

B. Position coordinates of atoms

We consider now a linear polymer chain with an appropriate degree of polymerization (e.g., DP=1000). We take as model a polyethylene chain. We thus have a homopolymer with a fixed valence angle θ between all the succeeding σ-bonds. These each have a rotational freedom which is characterized by the angles φ_i (Fig. 4). We start numbering the chain from one terminus in a way that we assign $i-3$ to the first atom, $i-2$ to the second atom, and so on. The purpose of the rather strange numbering at first sight will later turn out to be very efficient in leading to a clear presentation of the solution.

We now apply the geometric algebra as given by Hestenes and co-worker[4,5] to obtain position coordinates of the atoms of the main chain as function of the angles φ_i. For a polyethylene chain θ_i and l_i are constant and we thus denote them with θ and l. We later extend to the case where valence angles and bond lengths are not anymore constant along the main chain. A good example of such a situation is a polypeptide chain.

We start by laying down bonds $i-2$ and $i-1$ in a same plane (Fig. 5). On atom $i-2$ we define a dextral unit trivector (pseudoscalar) $\boldsymbol{\iota}=\boldsymbol{\sigma_1\sigma_2\sigma_3}=\boldsymbol{\sigma_1}\wedge\boldsymbol{\sigma_2}\wedge\boldsymbol{\sigma_3}$ which thus determines a local orthonormal coordinate system. We define the unit trivector in that way that $\boldsymbol{\sigma_1}$ has the same direction as bond λ_{i-2}. We choose $\boldsymbol{\sigma_2}$ to be parallel with bond λ_{i-1} (same direction). The relative orientation of bonds $i-2$ and $i-1$ with respect to $\boldsymbol{\iota}$ or vice versa is now fixed.

We now assume that bond λ_i lies in the plane parallel to the plane $\boldsymbol{\sigma_2\sigma_3}=\boldsymbol{\sigma_2}\wedge\boldsymbol{\sigma_3}$ as is shown in Fig. 5. This bond λ_i has a component $\lambda_{i\parallel}$ parallel with bond λ_{i-1} and another component $\lambda_{i\perp}$ perpendicular to λ_{i-1}. We easily see that $|\lambda_{i\parallel}|=-|\lambda_i|\cos(\theta)$ and $|\lambda_{i\perp}|=|\lambda_i|\sin(\theta)$. The perpendicular component $\hat{\lambda}_{i\perp}$ can now be written as

$$\hat{\lambda}_{i\perp}=-\boldsymbol{\iota}\boldsymbol{\sigma_1}\wedge\boldsymbol{\sigma_2},$$

which is equal to $\boldsymbol{\sigma_1}\times\boldsymbol{\sigma_2}$ as in classical vector algebra. Obviously, this can be rewritten as

$$\hat{\lambda}_{i\perp}=-\boldsymbol{\iota}\hat{\lambda}_{i-2}\wedge\hat{\lambda}_{i-1}.$$

We now have

$$\lambda_i=\lambda_{i\perp}+\lambda_{i\parallel}, \tag{8}$$

where $\lambda_{i\parallel}$ is equal to

$$\lambda_{i\parallel}=|\lambda_{i\parallel}|\hat{\lambda}_{i-1}=-l\cos(\theta)\hat{\lambda}_{i-1}. \tag{9}$$

In Eq. (8) $\lambda_{i\perp}$ is the variable component which introduces through the angle φ_i a rotation around the axis defined by bond $i-1$. In geometric algebra this rotation is described by use of a spinor[4] as follows:

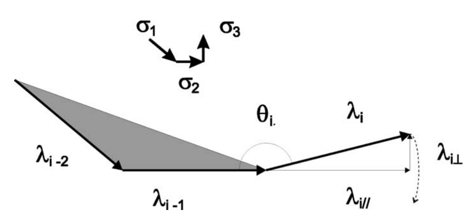

FIG. 5. Position of the bonds as described in the text for the derivation of the main expressions. The bonds λ_{i-2} and λ_{i-1} are in the same plane (filled area). The bond λ_i is the sum of a component parallel to λ_{i-1} ($\lambda_{i\parallel}$) and the perpendicular 1-vector $\lambda_{i\perp}$. The relation of the bonds with the orientation of the dextral unit trivector $\boldsymbol{\sigma_1\sigma_2\sigma_3}$ is also shown.

$$\lambda_{i\perp}(\varphi_i) = \mathbf{R}^\dagger \lambda_{i\perp} \mathbf{R},$$
$$\lambda_{i\perp}(\varphi_i) = l \sin(\theta) \mathbf{R}^\dagger \hat{\lambda}_{i\perp} \mathbf{R}. \tag{10}$$

The spinor $\mathbf{R}$ is a function of φ_i [thus $\mathbf{R}(\varphi_i)$]. The spinor is equal to

$$\mathbf{R} = e^{(1/2)\iota\varphi_i\hat{\lambda}_{i-1}}, \tag{11}$$

where the reverse spinor $\mathbf{R}^\dagger$ is easily determined as

$$\mathbf{R}^\dagger = e^{-(1/2)\iota\varphi_i\hat{\lambda}_{i-1}}.$$

If we substitute Eqs. (9) and (10) in Eq. (8) and change to operator notation we get

$$\lambda_i = l \sin(\theta) \mathcal{R} \hat{\lambda}_{i\perp} - l \cos(\theta) \hat{\lambda}_{i-1}. \tag{12}$$

In the derivation of Eq. (12) we assumed a constant length l and angle θ along the polymer chain. Equation (12) is recursive in nature and it is necessary to calculate first λ_i before being able to compute λ_{i+1}. As a consequence making l and θ variable does not introduce any problem to Eq. (12).

The more generalized equation, in which l and θ are made variable, is therefore

$$\lambda_i = - l_i \sin(\theta_i) \mathcal{R}(\iota\hat{\lambda}_{i-2} \wedge \hat{\lambda}_{i-1}) - l_i \cos(\theta_i) \hat{\lambda}_{i-1}. \tag{13}$$

This equation allows us to calculate by means of geometric algebra all the orientation 1-vectors of the chemical bonds of a main chain and consequently the position of all its atoms. In addition the geometry of each single bond (l_i, θ_i, and φ_i) can be freely chosen without any restriction.

We now derive the shortest possible representation of Eq. (13) for λ_i. We have for $\lambda_{i\perp}$ and $\lambda_{i\parallel}$ the following:

$$\lambda_{i\perp} = l_{i\perp}(-\iota\hat{\lambda}_{i-2} \wedge \hat{\lambda}_{i-1}) = -\iota l_{i\perp} \hat{\lambda}_{i-2} \wedge \hat{\lambda}_{i-1},$$

$$\lambda_{i\parallel} = l_{i\parallel} \hat{\lambda}_{i-1},$$

in which $l_{i\parallel} = -l_i \cos(\theta_i)$ and $l_{i\perp} = l_i \sin(\theta_i)$. We thus can write

$$\lambda_i = -\iota l_{i\perp} \hat{\lambda}_{i-2} \wedge \hat{\lambda}_{i-1} + l_{i\parallel} \hat{\lambda}_{i-1}. \tag{14}$$

More shortly, Eq. (14) can be written as

$$\lambda_i = \mathcal{O}_{i-2} \hat{\lambda}_{i-1}, \tag{15}$$

in which we define operator $\mathcal{O}_{i-2}$ as

$$\mathcal{O}_{i-2} \equiv (-\iota l_{i\perp} \hat{\lambda}_{i-2} \wedge + l_{i\parallel}).$$

Equation (14) represents only one orientation 1-vector, namely, that one we use to rotate it through an angle φ_i. We denote this specific 1-vector as

$$\lambda_i = \lambda_i(\varphi_i = 0) = \lambda_i^0.$$

When we derived Eq. (13) we first decomposed λ_i in a parallel and a perpendicular component after which we applied a rotation to the perpendicular component through the spinor operator $\mathcal{R}$. This, however, is not necessary since the spinor operator can work directly on the total λ_i^0 1-vector, which yields

$$\lambda_i(\varphi_i) = \mathcal{R}\lambda_i^0 = \mathbf{R}^\dagger \lambda_i^0 \mathbf{R}. \tag{16}$$

This is the shortest possible equation (canonical form) which describes the resulting orientation 1-vector. If we combine this with Eq. (15) we get

$$\lambda_i(\varphi_i) = \mathcal{R}(\varphi_i) \mathcal{O}_{i-2} \hat{\lambda}_{i-1}. \tag{17}$$

The next bond λ_{i+1} is then to be represented as

$$\lambda_{i+1}(\varphi_{i+1}) = \mathcal{R}(\varphi_{i+1}) \mathcal{O}_{i-1} [(1/l_i) \mathcal{R}(\varphi_i) \mathcal{O}_{i-2} \hat{\lambda}_{i-1}], \tag{18}$$

in which the total operator product is noncommutative. Consequently, it is not possible to collect the like terms such as the spinor operators. Such a possible reduction would make the representation and calculation a lot easier. Indeed, the spinor product can then be converted to one new spinor. Further in the text we will derive an equation where this is possible (case of rotating chain segments).

The effect of $\mathcal{R}$ on the parallel component can be seen as an identity multiplication which keeps $\lambda_{i\parallel}$ invariant under the rotation. If we look at Eq. (16) we observe a very elegant and simple equation to make the calculation. Elaboration would yield Eq. (13). Working out this latter equation would give a much larger expression which is not practical in use. To benefit further from Eq. (13) we apply the second line of the under Eq. (2). We can herewith convert the outer product to a geometrical product and in doing this we express the 1-vectors also in their components. Secondly, we bring the spinor in its nonexponential form. The resulting equation only consists of sums and products of the unit multivectors which can then be computed much more easily.

The orientation 1-vectors of the chemical bonds can easily be related to the position coordinates of the atoms (Fig. 6) and for the position coordinate $\mathbf{p}_n$ of the nth atom we get

$$\mathbf{p}_n = \sum_{i=2}^{n} \lambda_i, \tag{19}$$

in which we assume that the first atom of the polymer is numbered 1 and that its position is the origin. Of course atom 1 has no λ_1 vector and as a consequence the index i starts running from 2.

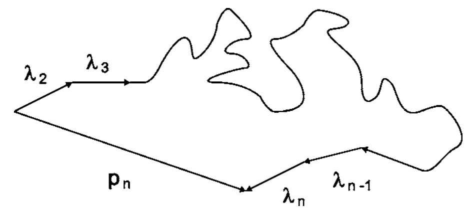

FIG. 6. The position coordinate $\mathbf{p}_n$ of atom n in the main chain. This coordinate is the sum of all the bond vectors λ_i up to atom n. Each of those λ_i is computed by geometric algebra. The first bond is λ_2 since we started numbering the first atom as 1: It is the bond between atoms 1 and 2 according to our conventions.

C. Branched polymers

In the foregoing we looked at a linear polymer chain (polyethylene). The approach is, however, not only restricted to the linear case; branched polymers can be described by a minor extension of what has been presented. If we consider the same polyethylene chain, we can imagine that one of the hydrogen atoms is substituted for an aliphatic carbon side chain (e.g., dodecyl group). The length of the side chain will mainly determine whether the total structure can still be considered as a linear polymer or is rather a true branched polymer. If the branch point is main chain atom i, bond $(i+1)_{\mathrm{MC}}$ of the main chain and the one of the side chain $(i+1)_{\mathrm{SC}}$ will clearly differ whereas bond i will still be joint. The relation between $(i+1)_{\mathrm{MC}}$ and $(i+1)_{\mathrm{SC}}$ is now determined by the chemical nature (hybridization of atom i) of the atom where the branching occurs. Since the bond i is common to both branches the 1-vector $\hat{\lambda}_i$ is also common in their spinors. As a result only the angle φ can differ in the spinor. If we look along the axis of bond i in the direction of the branching chains the bonds will be rotated by a fixed angle $\Delta\varphi$ with respect to each other. In our example (sp^3-hybridization) the three bonds are separated by an angle of 120°. It ensues that their spinors will also differ by this value in angle and as a result we have

$$\mathcal{R}_1 = \mathcal{R}(\varphi),$$

$$\mathcal{R}_2 = \mathcal{R}(\varphi + 120°),$$

$$\mathcal{R}_3 = \mathcal{R}(\varphi + 240°).$$

In our example the side chain will be described by spinor $\mathcal{R}_2$ or $\mathcal{R}_3$ if $\mathcal{R}_1$ is considered as the main chain spinor. In case of sp^2-hybridization we will have for the side chain

$$\mathcal{R}_{\text{side chain}} = \mathcal{R}(\varphi + 180°).$$

For the branching chain segment we thus calculate up to atom i the same series of spinors as for the main chain. Then we calculate with the first side chain spinor and proceed with a new series of spinors defining the side chain. The latter are completely independent of spinors of the main chain conformation after the branch point.

The theoretical approach to branching does thus not introduce much difficulty. Of course the presence of branches will result in additional calculations (e.g., in dendrimers). We mention that a hydrogen atom can be considered as a branch or side chain and therefore its coordinates can be calculated in this way.

D. Rotation of chain segments

The normalized 1-vector $\hat{\lambda}_i$ can be represented in terms of the involved position coordinates $\mathbf{p}_{i-1}$ and $\mathbf{p}_i$ as follows:

$$\hat{\lambda}_i = \frac{\mathbf{p}_i - \mathbf{p}_{i-1}}{|\mathbf{p}_i - \mathbf{p}_{i-1}|} = \hat{\Delta}\mathbf{p}_{i(i-1)}, \tag{20}$$

with $\hat{\boldsymbol{\Delta}}\mathbf{p}_{i(i-1)}$ as alternative notation for $\hat{\lambda}_{i-1}$. More generally we define the displacement 1-vector $\boldsymbol{\Delta}\mathbf{p}_{ji}$ which is equal to $\mathbf{p}_j - \mathbf{p}_i$ with $j > i$ and integer valued. Besides the case where

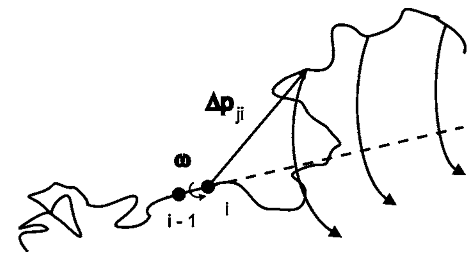

FIG. 7. Representation of the action of spinor operator $\mathcal{R}$ on a chain segment by rotating this segment around the bond connecting atoms $i-1$ and i. The terminal segment is rotated in a clockwise sense through an angle of ω rad. The spinor operator is $\mathcal{R}(\omega)$ and the vectors $\boldsymbol{\Delta}\mathbf{p}_{(i+\Delta i)i}$—with $\Delta i \geq 1$ and integer—need to be known to calculate the new coordinates of the terminal segment.

$j = i+1$, the 1-vector $\boldsymbol{\Delta}\mathbf{p}_{ji}$ does not anymore correspond physically to a bond (e.g., Fig. 7). As start situation we assume that the chain is in an initial conformation defined by the set $\{l_i, \theta_i, \varphi_i\}$. The original position coordinates are now given by $\mathbf{p}_i$ and an atom j has thus coordinate $\mathbf{p}_j$ in space. We consider now the rotation through an additional angle ω of atom j around bond-axis $(i-1 \to i)$ yielding

$$\boldsymbol{\Delta}\mathbf{p}'_{ji} = \mathcal{R}\boldsymbol{\Delta}\mathbf{p}_{ji},$$

in which $\mathcal{R}$ encloses the spinor $\mathbf{R}$ which is equal to

$$\mathbf{R} = e^{(1/2)\iota\omega\hat{\lambda}_i}.$$

This yields for the new position coordinate $\mathbf{p}'_j$ as follows:

$$\mathbf{p}'_j = \mathbf{p}_i + \mathcal{R}\Delta\mathbf{p}_{ji} \tag{21}$$

and if the conformation of the atoms with index $j > i$ was already known this equation should be applied to each of the $\boldsymbol{\Delta}\mathbf{p}_{ji}$. Hence, the total chain segment from atom i will be rotated through an angle ω (Fig. 7). It should be remarked that the new conformation can also be calculated in another way. If $\mathcal{R}(\varphi_{i+1})$ is the original spinor operator of the involved bond (note the functional use of φ_{i+1} corresponding with rotation around physical bond i) which will rotate the chain segment, the new spinor operator $\mathcal{R}(\varphi_{i+1}+\omega)$ is necessary to generate the new structure. In that case the original spinors $\mathbf{R}$ for $j > i+1$ remain valid and should be applied in the calculation. The difference lies in the fact that the latter method needs the parameters of the structure while the approach along the outline in this section only needs the actual conformation (position coordinates). The difference can be described as operating on a conformation [Eq. (21)] or generating a new structure. Both approaches can be useful depending on the scope and the type of structural data available. As an example we mention the data present in pdb files. That type of file contains the position coordinates and not the set $\{l_i, \theta_i, \varphi_i\}$. In that case Eq. (21) should be used. On the contrary if we have an α-helical segment for which we know the geometrical parameters the second approach is the preferable one.

Let us now assume that a chain has a conformation determined by the set $\{l_i, \theta_i, \varphi_i\}$. Suppose that a rotation occurs around bond i which gives $\mathbf{p}'_j$ as described by Eq. (21). For another atom k for which $k > j$ the same rotation gives

$$\mathbf{p}'_k = \mathbf{p}_i + \mathcal{R}\Delta\mathbf{p}_{ki}, \tag{22}$$

which is also equal to

$$\mathbf{p}'_k = \mathbf{p}'_j + \Delta\mathbf{p}'_{kj}.$$

On substituting for $\mathbf{p}'_j$ we get

$$\mathbf{p}'_k = \mathbf{p}_i + \mathcal{R}\Delta\mathbf{p}_{ji} + \Delta\mathbf{p}'_{kj},$$

and this yields the following for $\Delta\mathbf{p}'_{kj}$ by use of Eq. (22):

$$\Delta\mathbf{p}'_{kj} = \mathcal{R}_i(\Delta\mathbf{p}_{ki} - \Delta\mathbf{p}_{ji}) = \mathcal{R}_i\Delta\mathbf{p}_{kj}, \tag{23}$$

in which the subscript i of $\mathcal{R}_i$ refers to the spinor acting along bond (axis) i. If now the adapted structure undergoes a second rotation around bond j we get the following as new coordinate for atom k by use of Eqs. (21) and (23):

$$\mathbf{p}''_k = \mathbf{p}'_j + \mathcal{R}_j\mathcal{R}_i\Delta\mathbf{p}_{kj}.$$

After substitution of Eq. (21) this yields

$$\mathbf{p}''_k = \mathbf{p}_i + \mathcal{R}_i\Delta\mathbf{p}_{ji} + \mathcal{R}_j\mathcal{R}_i\Delta\mathbf{p}_{kj}.$$

In case of n successive rotations an arbitrary atom k of the chain will have as coordinate

$$\mathbf{p}^{(n)}_k = \mathcal{R}_0\mathbf{p}_i + \mathcal{R}_{ji}\Delta\mathbf{p}_{ji} + \mathcal{R}_{kj}\mathcal{R}_{ji}\Delta\mathbf{p}_{kj} + \cdots + \mathcal{R}_{nm}\mathcal{R}_{ml}\cdots\mathcal{R}_{ji}\Delta\mathbf{p}_{nm},$$

with $i<j<k<\cdots<m<n$. The subscript i of the involved spinor operators indicates the bond about which the rotation is performed while the subscript j refers to the atom up to which the involved spinor operators act.

An alternative notation is

$$\mathbf{p}^{(n)}_k = \mathcal{R}_0\Delta\mathbf{p}_0 + \mathcal{R}_1\Delta\mathbf{p}_1 + \mathcal{R}_2\mathcal{R}_1\Delta\mathbf{p}_2 + \cdots + \mathcal{R}_n\mathcal{R}_{n-1}\cdots\mathcal{R}_1\Delta\mathbf{p}_n, \tag{24}$$

in which $\mathcal{R}_0 \equiv \mathcal{E}$ which acts upon $\Delta\mathbf{p}_0 = \mathbf{p}_1$ (identity operation on position 1-vector). The subscripts of the spinors in Eq. (24) indicate the order of the segments along the chain in which the rotations occur; the displacement 1-vectors $\Delta\mathbf{p}$ refer to the segments upon which these spinors act and also indicate the relative position of the segments along the chain. The significance of Eq. (24) can thus be summarized as follows: A polymer chain was in an initial conformation. Up to atom $\mathbf{p}_1$ the chain conformation is unaltered, the segment starting at $\mathbf{p}_1$ and ending at atom with position $\mathbf{p}_1+\Delta\mathbf{p}_1$ is rotated through an angle ω_1 (by spinor operator $\mathcal{R}_1$). The following segment is then rotated two times and this corresponds to the action of the operator $\mathcal{R}_2\mathcal{R}_1$ (Fig. 8). This continues till all the rotations are performed.

Equation (24) can still be made shorter,

$$\mathbf{p}^{(n)}_k = \sum_{i=0}^{n}\left[\left(\prod_{j=0}^{i}\mathcal{R}_j\right)\Delta\mathbf{p}_i\right] = \sum_{i=0}^{n}\mathcal{R}^{\pi}_i\Delta\mathbf{p}_i, \tag{25}$$

in which $\mathcal{R}^{\pi}_i$ represents the resulting spinor which is thus equal to

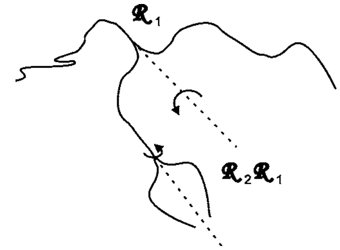

FIG. 8. Schematic illustration of successive rotations of polymer chain segments by means of spinor operators. In the figure the spinor operators $\mathcal{R}_1$ and $\mathcal{R}_2$ act on the main chain. We can see that the first segment is unaltered while the terminal segment is rotated twice. The last rotation can be calculated by using the spinor operator product $\mathcal{R}_2\mathcal{R}_1$ which is actually a new spinor.

$$\prod_{j=0}^{i}\mathcal{R}_j = \mathcal{R}^{\pi}_i.$$

The right side spinor is not simply a symbolic conversion of the involved product but a real new spinor which can be calculated. The superscript π refers to the fact that it is a product of spinors. In contrast, in the case of the bond calculations, a nonspinor operator $\mathcal{O}$ works first on λ^0_i 1-vector and subsequently alters this vector. Therefore we have to respect the order and noncommutativity of the different operators in these equations and hence no collection of spinors and subsequent reduction to one spinor is possible.

Equation (25) allows us to calculate the position of atom k in chain segment n ($i^{(n)}$ refers to the index of segments and $\Delta\mathbf{p}_{i=n}$ ends at atom k) on which the nth rotation is performed. If we cancel the last (canonical) term the lower position coordinates can be calculated by the remaining equation. Continuing in this way we can obtain all atom coordinates. In contrast with Eq. (13), which is the formula to generate a structure, Eq. (25) can be understood to operate on a chain structure and modify it.

E. Advantages of the method

The presented method is very suitable for molecular modeling since it allows us to easily generate a chemical structure given its set of parameters $\{l_i, \theta_i, \varphi_i\}$. The rotation of chain segments is described in a simple way and conformational changes can be computed accordingly. This method should in the first place be regarded as a powerful computational tool. As presented here it does, however, not allow the prediction of the conformation of a chemical structure based on energetic considerations.

A comparison with matrix algebra, which is an excellent mathematical framework for calculation and analysis of polymer conformations, is appropriate to evaluate some features concerning computation by means of geometric algebra. Firstly, geometric algebra makes no use of reference frames. In this paper we only use the dextral unit trivector ι for the construction of the 1-vector $\lambda_{i\perp}$ by analogy with the previous bond 1-vectors (Sec. III B). This should be mentioned as to avoid confusion. In matrix algebra, however, reference frames are necessary to calculate the actual orientation of bond vectors in space. This is done by squared transformation matrices and cannot be skirted around as follows:[17]

$$\lambda_{i;1} = [{}^{1}\mathbf{T}_2 \cdot {}^{2}\mathbf{T}_3 \cdot \ \ldots \ \cdot {}^{i-2}\mathbf{T}_{i-1} \cdot {}^{i-1}\mathbf{T}_i] \cdot \lambda_{i;i}, \tag{26}$$

in which

$$\lambda_{i;i} = \begin{bmatrix} l_i \\ 0 \\ 0 \end{bmatrix}. \tag{27}$$

Equation (26) can also be rewritten as

$$\lambda_{i;1} = [{}^{1}\mathbf{T}_{i-1} \cdot {}^{i-1}\mathbf{T}_i] \cdot \lambda_{i;i}, \tag{28}$$

in which ${}^{1}\mathbf{T}_{i-1}$ is the product $\Pi_1^{i-2}\, {}^{k}\mathbf{T}_{k+1}$. In Eq. (26), ${}^{i-1}\mathbf{T}_i$ is the $\mathfrak{R}^{3\times 3}$ matrix which transforms the coordinates of λ_i in reference frame i $(\lambda_{i;i})$ to the corresponding coordinates in frame $i-1$. By successive multiplication with the transformation matrices ${}^{i-1}\mathbf{T}_i$ of the previous reference frames we finally obtain the coordinates in reference frame 1 $(\lambda_{i;1})$. This means that matrix algebra calculates—as to say—backward while geometric algebra calculates bond 1-vectors forward. It is important here to notice that it is mainly the calculation of the bond vectors that differs in the two approaches. The final position coordinates are obtained in the same way (by addition of the bond vectors). As shown in Eq. (26) it is crucial in matrix algebra that each reference frame is related to its predecessor frame. Thus, all the transformation matrices need to be known to calculate the absolute orientation of a certain bond vector in space. In geometric algebra, on the other hand, for the calculation of the absolute orientation of a bond vector only the two previous bond vectors are needed:

$$\lambda_i(\varphi_i) = \mathcal{R}(\varphi_i)\mathcal{O}_{i-2}\hat{\lambda}_{i-1}. \tag{29}$$

The information on bond λ_{i-2} in Eq. (29) is contained in operator $\mathcal{O}_{i-2}$. Geometric algebra thus calculates forward.

A second difference is the fact that in geometric algebra the symbolic representation of the equations refers more transparently to the operations of the equations and their involved variables. This is primarily the result of the use of spinors. The axis of rotation $\hat{\lambda}_{i-1}$ and dihedral angle φ_i are clearly represented in the spinor equation as separate variables:

$$\mathbf{R} = e^{(1/2)\iota\varphi_i\hat{\lambda}_{i-1}}. \tag{30}$$

This is not the case in matrix algebra where such a straightforward relation with the transformation matrix ${}^{i-1}\mathbf{T}_i$ does not exist. The transformation matrix which is of importance is[7]

$${}^{i-1}\mathbf{T}_i(\theta_i,\varphi_i) = \begin{bmatrix} \cos\theta_i & \sin\theta_i & 0 \\ -\sin\theta_i\cos\varphi_i & \cos\theta_i\cos\varphi_i & \sin\varphi_i \\ \sin\theta_i\sin\varphi_i & -\cos\theta_i\sin\varphi_i & \cos\varphi_i \end{bmatrix}. \tag{31}$$

In matrix ${}^{i-1}\mathbf{T}_i(\theta_i,\varphi_i)$, θ_i and φ_i are, respectively, the valence and dihedral angles related to bond i. The axis of rotation is not represented in the matrix ${}^{i-1}\mathbf{T}_i$.

Furthermore geometric algebra is computationally more efficient than matrix algebra. This feature of geometric algebra was already proven in 1970 by Ickes.[18] He established this fact by counting and comparing elementary operations needed to compose rotations on a computer. More recently similar studies have been done in research fields of robotics and graphical sciences.[19–21] They showed a clear advantage for geometric algebra. However, in our case we must first construct a nonrotated virtual bond vector λ_i^0 and this results in additional operations. As a result, an independent analysis in this paper is necessary to compare the computational efficiency of matrix and geometric algebra. We will therefore examine now in more detail the number of elementary operations in the calculation of a final bond vector.

In matrix algebra this calculation involves the choice of coordinates in the local frame, the construction of a transformation matrix ${}^{i-1}\mathbf{T}_i$ which relates the local frame to the predecessor frame, and the knowledge of the transformation matrix which relates the predecessor frame to the first frame, ${}^{1}\mathbf{T}_{i-1}$. The storage of the latter matrix requires nine memory allocations (9^M). The construction of the local transformation matrix requires four multiplications $(4^\times)$ and storage of four memory allocations. The construction of the local coordinates requires storage of the length of the bond vector but this is necessary as well in geometric algebra. The multiplication of the two matrices requires 9×3 multiplications $(27^\times)$ and 9×2 additions (18^+). From these numbers three multiplications and three additions have to be subtracted due to the presence of a zero element t_{13} in the local transformation matrix. Finally, the multiplication of the resulting matrix with the local coordinates includes a further three multiplications $(3^\times)$. This yields in total $(4+27+3-3)^\times=31^\times$, $(18-3)^+=15^+$ and $(9+4)^M=13^M$. It is important to mention that due to the clever choice of the local coordinates an additional $(3+3)^\times=6^\times$ and $(3+3)^+=6^+$ are avoided in matrix algebra. Otherwise the total number of operations would be $37^\times$, 21^+, and 13^M. Geometric algebra requires construction of two spinors. This needs $3^\times$ and 3^M. By a further $3^\times$ the reverse spinor is constructed. The construction of the perpendicular component requires $6^\times$ and 3^+ (calculation of outer product). The total unit bond vector that will be rotated requires $(3+3)^\times=6^\times$ and 3^+. Multiplication $(3^\times)$ yields the bond vector. The spinor operator product requires $24^\times$ and 12^+. The equivalent quaternion product, in which $\mathbf{q}=(s,\mathbf{v})$ and $\mathbf{p}=(w,\mathbf{x})$ are quaternions with a scalar and vector part, illustrates this more clearly,[22]

$$\mathbf{qpq}^{-1} = \left(w, \frac{1}{|\mathbf{q}|^2}((s^2-\mathbf{v}\cdot\mathbf{v})\mathbf{x} + 2(\mathbf{x}\cdot\mathbf{v})\mathbf{v} - 2s(\mathbf{x}\times\mathbf{v}))\right). \tag{32}$$

In Eq. (32), $\mathbf{p}$ is equivalent to the unit bond vector we construct whereas quaternion $\mathbf{q}$ is equivalent to spinor $\mathbf{R}$ and $\mathbf{q}^{-1}$ equivalent to $\mathbf{R}^\dagger$. The scalar part w is of no importance and $|\mathbf{q}|=1$ because we work with unitary spinors. In total this yields for geometric algebra: $(3+3+6+6+3+24)^\times=45^\times$, $(3+3+12)^+=18^+$ and 3^M.

TABLE I. Number of elementary operations in matrix and geometric algebra for the calculation of a bond vector. The total number is nonweighed and underestimates the relative importance of memory access. The number of operations, respectively, unrelated (first number) and related to the rotation itself, is represented in parentheses.

	Elementary operations			
	$\times$	$+$	Memory	Total
Matrix algebra	31 (4+27)	15	13	59
Geometric algebra	45 (21+24)	18	3	66

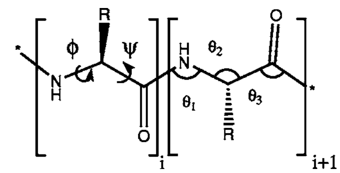

FIG. 9. Peptide units i and $i+1$ in a polypeptide backbone. The side chain R determines the nature of each peptide unit in the chain. The dihedral angles ϕ and ψ are shown in peptide unit i. The stereochemistry of bond CO–NH is fixed and there is no variable dihedral angle. On the right (peptide unit $i+1$) we see the three involved valence angles (θ_1, θ_2, and θ_3) which are constant (neglecting vibrations of the bonds).

From our analysis (Table I) we conclude that more multiplications are needed in geometric algebra but much more memory allocations in matrix algebra. If all operations are added we obtain for matrix algebra in the most ideal case: $31^{\times}+15^{+}+13^{M}=59$ operations while in the case of geometric algebra: $45^{\times}+18^{+}+3^{M}=66$ operations. Considering the fact that memory access costs normally a larger amount of time than additions or multiplications,[20] the computational potential of matrix and geometric algebra seems comparable. In the nonideal case for matrix algebra the total number of operations needed is 71. In that case, due to the memory cost, geometric algebra is clearly preferable. Matrix algebra, in the ideal case, has similar computational efficiency as geometric algebra because of the fact that the local reference frame can be chosen optimally. For geometric algebra the inefficient part is situated in the construction of the unit bond vector that will be rotated. If this bond vector would be available at first hand the total number of operations would reduce to $24^{\times}+3^{\times}+12^{+}+3^{M}=42$ operations and, given the low storage cost, this would yield a clearly better performance than with matrix algebra. Based on this analysis it can be concluded that geometric algebra is at least equally efficient as matrix algebra. The superior efficiency as in performing only rotations is reduced due to the construction of the nonrotated virtual bond vector which will be rotated. It may, however, not be forgotten that the computation of this vector makes this method reference-frame-free which is at the same time an important strength of the outlined approach.

In conclusion, the only restriction at present for the introduction of geometric algebra is the fact that it needs further development and incorporation of methods which allow prediction concerning the energy of conformations but this should be perfectly possible. It was also our choice to only introduce the mathematical framework in this paper. Due to its computational possibilities, geometric algebra has the potential to establish itself firmly for modeling purposes.

In Appendix A, we compare in more detail computational aspects of matrix and geometric algebra. Furthermore Appendix B discusses the conversion between these two algebras.

IV. EXAMPLE

We applied the presented method to a specific chemical structure. For the calculations we made use of MATLAB 5.3,[23] whereas for visualization both MATLAB 5.3 and PYMOL (Ref. 24) were used. We opted for the use of a protein as a test case. Proteins are made up of peptide units which each are characterized by three different bond lengths, three different valence angles, and two variable dihedral angles (φ and ψ). The third dihedral angle (around CO–NH) is fixed (180°) because of the peptide plane (Fig. 9).

As a result of the nature of this peptide unit and of the folding of the polypeptide chain, in general a highly complex and compact globular structure is obtained. We chose the monomer unit of oxymyoglobin (Fig. 10, left image) both because of the fact that it is well known and because of the crystallographic resolution which is 1.00 Å. We retrieved the pdb file (1A6M.pdb) from the RCSB Protein Data Bank. The oxymyoglobin monomer unit is a low-molecular mass polypeptide chain (M_r=17 054) which is composed of 151 amino acid residues. As can be seen in Fig. 10 (left image) it is globular in structure and has a considerable helical content.

Practically we worked as follows. After having retrieved the pdf file we made in PYMOL a selection of the main chain atoms and saved this selection. The Windows program TextPad (freeware) was then used to get the data of the separate columns of the x-, y-, and z-coordinates. This means that all the original coordinates as determined by crystallographic methods were at our disposal for the computations in MATLAB. In MATLAB several small self-written m-files were used to calculate the bond lengths, valence, and dihedral angles out of these data (see Appendix C).

The most difficult parameter to calculate, the dihedral angle, was obtained as follows: Two consecutive vector

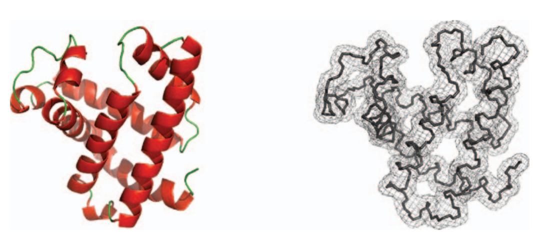

FIG. 10. (Color) On the left a cartoon image of the main chain of a monomer unit of oxymyoglobin. It shows the main secondary structural elements and their relative orientation as in the native conformation. Image created by use of PYMOL with the pdb file of oxymyoglobin (1A6M). On the right we see the backbone structure as calculated by geometric algebra. To make the image, the results of the calculations in MATLAB were written to an xyz-file which is supported by PYMOL. The image was then rendered by PYMOL as sticks surrounded by a mesh surface. The orientation in space is similar as to make the comparison easier.

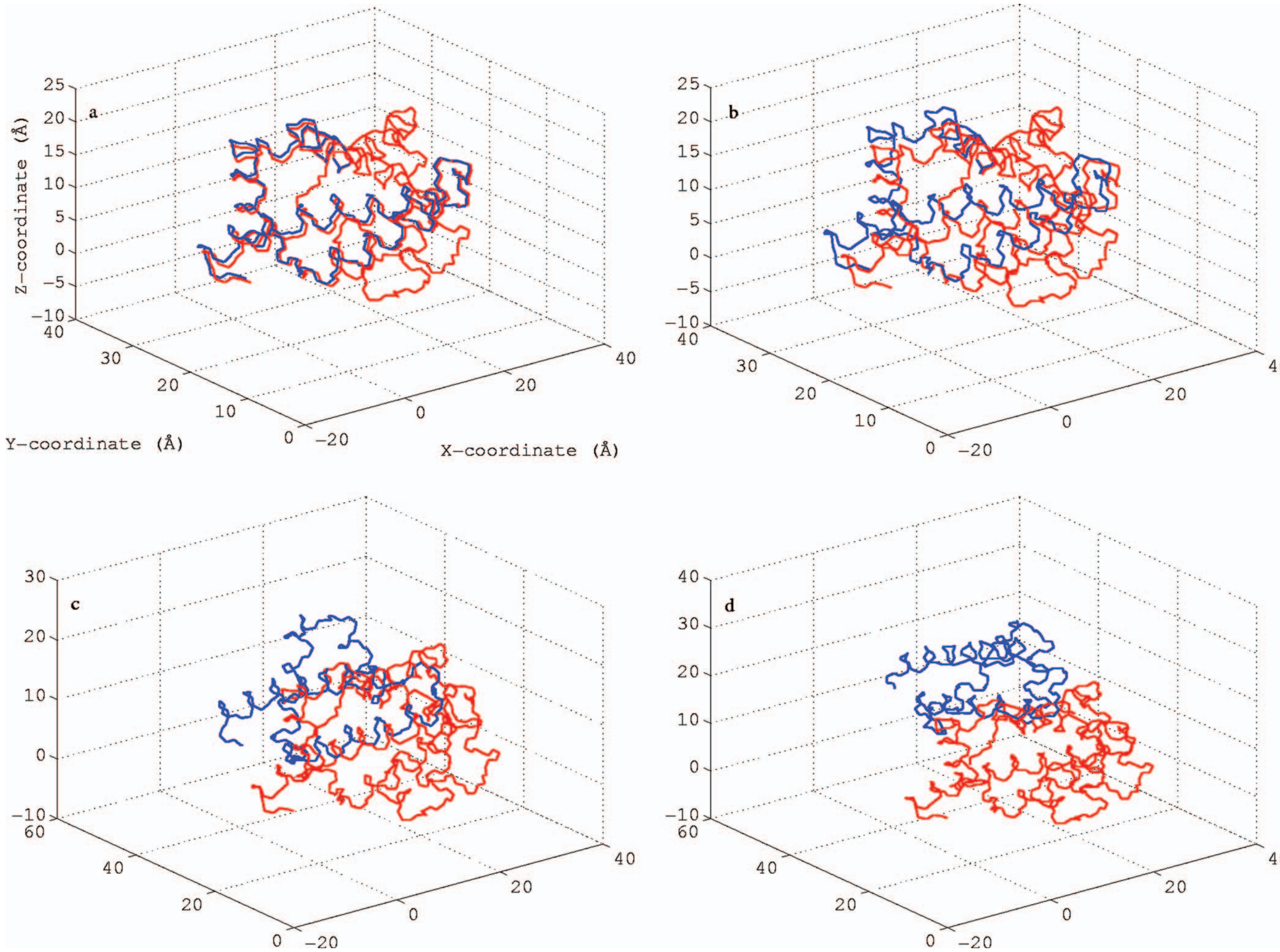

FIG. 11. (Color) Illustration of rotations of a terminal chain segment of oxymyoglobin. Rotations of 2.5° (a), 10° (b), 40° (c), and 90° (d) were performed around bond 200 of the main chain. The original conformation is given in red, and the rotated segment is each time drawn in blue. Figure made with MATLAB.

products of bonds in the chain (e.g., product for bond 1 and 2+product for bond 2 and 3) were calculated, yielding two vectors whose scalar product allows us to calculate the dihedral angle since the computed vectors of the vector products lie in parallel planes. An extra algorithm in MATLAB was necessary to calculate the correct value of the dihedral angle since there is no discrimination between an angle φ and $2\pi-\varphi$. No use was made of geometric algebra in all those steps. Once this was done we calculated by means of Eqs. (13) and (19) the position coordinates (Fig. 10, right side). We then aligned those position coordinates with the original ones from the pdb file and calculated the root-mean-square deviation. This calculated value was found to be 3.55×10^{-13} pm. For comparison, the length of a classical N–C_α bond in a polypeptide chain is approximately 145 pm.

We also made a calculation on the basis of Eq. (25) for the rotation of chain segments. We continued working with oxymyoglobin. The result of a few of such calculations is shown in Fig. 11. Different rotations were performed around the bond axis (atom 199$\rightarrow$atom 200) with increasing dihedral angles. These correspond to rotations around the NH–CH bond (see Fig. 9) of amino acid 67 in the polypeptide sequence of oxymyoglobin.

V. CONCLUSIONS

In this paper we presented a method which makes use of geometric algebra to calculate polymer conformations. With this method we calculated the position coordinates of the main chain atoms. Moreover, we made an extension for side chains (branched polymers) and described rotation of chain segments. This allows the generation of conformations of macromolecular structures and their manipulation for modeling purposes. We used the monomer unit of oxymyoglobin as an example for such a calculation. In principle any type of chemical structure can be created, e.g., also small organic molecules.

The applicability is mainly in the area of modeling with emphasis on the computational aspects of the method. We did not consider in this paper the broad topic of estimating the energies of conformations since the primary goal was to introduce geometric algebra as a new alternative for calculations in the world of modeling. This is justified because geometric algebra (i) allows us to work reference-frame-free, (ii) has a superior symbolic notation to matrix algebra, and (iii) has a computational efficiency which is at least equal to that of matrix algebra.

ACKNOWLEDGMENTS

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APPENDIX A: FURTHER ANALYSIS OF COMPUTATIONAL ASPECTS

Suppose we want to assign a new conformation to a large terminal segment of a given polymer chain conformation, having the coordinates $\mathbf{p}_i$, unit bond vectors $\hat{\lambda}_i$, and the transformation matrices ${}^{1}\mathbf{T}_{i-1}$ at our disposal. In both matrix and geometric algebra the normal computational schemes of both methods are applied from the first altered bond. A fundamental difference in matrix algebra compared to geometric algebra is that ${}^{1}\mathbf{T}_{i-1}$ needs to be known. When one has generated a chain model, this matrix will be available because of the computations and no problem arises. However, if only position coordinates are known, the matrix ${}^{1}\mathbf{T}_{i-1}$ has to be calculated and this demands extra computations which are avoided in geometric algebra. In the latter case geometric algebra clearly has the edge.

Let us consider now that we change the conformation in the direction of the origin of the chain. In geometric algebra this is done by reversing the sense of the bond vectors ($-\lambda_{i+2}$ and $-\lambda_{i+1}$) and changing the order of the bonds upon which $\mathcal{O}_{i-2}$ acts. For the remainder the normal scheme is followed. In matrix algebra the situation is much more complex since the first bond is our reference frame. Keeping that bond as a reference frame would imply that the whole chain has to be recalculated. A better solution is to choose the first unaltered bond as a new reference frame. The reverse orientation of this bond is now taken as a positive axis. We then calculate with the normal procedure the new bond vectors of the involved segment along positive coordinates. After this, the signs in all three separate dimensions of the orientation coordinates are made opposite (executing a point reflection). At this stage the new conformation can be calculated but not yet with the correct orientation in space. Therefore, you still need the knowledge of ${}^{1}\mathbf{T}_{i-1}$ after which it is possible to calculate the correct position coordinates. Careful analysis is thus needed to continue calculations and the approach is not straightforward. Hence, geometric algebra is clearly superior since the method can be applied directly without any further analysis and with the same computational scheme.

In case a large segment—with fixed conformation—is rotated, in matrix algebra different approaches are possible. The most logical solution is the one in which use is made of matrix ${}^{1}\mathbf{T}_{i-1}$ and the unaltered matrices ${}^{i-1}\mathbf{T}_i$. At the altered bond, however, a new ${}^{i-1}\mathbf{T}_i$ is created. The remainder of the calculations is the same as in the normal scheme. In geometric algebra, on the other hand, the difference vectors $\mathbf{\Delta p}_{ji}$ are needed. These are then rotated by the involved spinor and yield the new $\mathbf{\Delta p}'_{ji}$. With these vectors the bond vectors can be computed by subtraction.

We can conclude that if conformations are changed in the reverse sense of the chain construction or if large fixed segments are rotated, geometric algebra is the better approach. In case only position coordinates are available, geometric algebra is even more preferable. Moreover, geometric algebra has the extra advantages of a clear notation and to be reference-frame-free.

APPENDIX B: CONVERSION BETWEEN GEOMETRIC ALGEBRA AND MATRIX ALGEBRA

Although geometric algebra is preferable, large segments of a fixed conformation can also be rotated with matrix algebra. Therefore conversion between the two methods is useful. In Ref. 4 some useful equations are derived. To construct the transformation matrix when a spinor is given we use

$$t_{jk} = (2\alpha^2 - 1)\delta_{jk} + 2\beta_j\beta_k - 2\alpha\left(\sum_i \beta_i \epsilon_{ijk}\right), \tag{B1}$$

where δ_{jk} is the Kronecker delta and $\epsilon_{ijk} \equiv -\langle \iota\sigma_i\sigma_j\sigma_k\rangle_0 = \sigma_i\cdot(\sigma_j\times\sigma_k)$ the Levi-Civita symbol. The α and β_j are the components of spinor $\mathbf{R}=\alpha+\iota\beta$ and $\alpha=\cos(\varphi/2)$ and $\beta = \sin(\varphi/2)\hat{\lambda}$. It is appropriate to work almost fully in the geometric algebra framework except for the rotation of large unaltered chain segments. Hence, Eq. (B1), together with the general spinor equation (15), composes a good computational scheme. Nevertheless it is useful to also include the equations for conversion from matrix algebra to geometric algebra.

A spinor $\mathbf{R}$ can be constructed from a transformation matrix ${}^{i-1}\mathbf{T}_i(\equiv {}^{i-1}\mathbf{S}_i)$ by

$$\pm\mathbf{R} = \frac{1+\mathbf{S}}{|1+\mathbf{S}|} = \frac{1+\mathbf{S}}{2\langle 1+\mathbf{S}\rangle_0^{1/2}}, \tag{B2}$$

in which $\mathbf{S}$ is a spinor constructed by means of matrix elements s_{kj} by the following equation:

$$\mathbf{S} = \sum_j \sum_k s_{jk}\sigma_j\sigma_k \tag{B3}$$

with running indices j and k from 1 to 3. The drawback of Eq. (B2) is that it is undefined in case of a dihedral angle $\varphi=180°$ whence $\alpha=\langle\mathbf{R}\rangle_0=0$ and $\mathbf{S}=-1$. In that case the following set of equations is necessary to use:

$$\beta = (1+s_{33})^{-1}[\beta_3(\boldsymbol{\sigma}_3+\mathbf{e}_3) + \alpha\boldsymbol{\sigma}_3\times\mathbf{e}_3], \tag{B4}$$

$$\alpha = \frac{s_{12}-s_{21}}{4\beta_3}, \tag{B5}$$

$$4\beta_3^2 = 1 + s_{33} - s_{11} - s_{22}. \tag{B6}$$

In this set of equations one has to calculate first β_3, subsequently α after which it is possible to calculate β. This scheme thus gives both α an β which compose the entire spinor. In general it is, however, preferential to proceed only from geometric to matrix algebra since a general applicable formula is present.

TABLE II. Main m-file chain.m and auxiliary files which form the basis of the self-written program. The file chain.m contains subroutines which refer to the auxiliary files. These help files take arguments (e.g., **a** and **b** or **B** and **T**) of which the form depends on the function of the help file itself.

m-file	Function of the file
chain.m	Main program file
geom11(a,b).m	Computes the total geometric product **ab** of 1-vectors **a** and **b**
geom11in(a,b).m	Computes the inner product $\mathbf{a}\cdot\mathbf{b}$ of 1-vectors **a** and **b**
geom11out(a,b).m	Computes the outer product $\mathbf{a}\wedge\mathbf{b}$ of 1-vectors **a** and **b**
geomi1(b).m	Computes the geometric product $\imath\mathbf{b}$ with **b** a 1-vector
geomi2(B).m	Computes the geometric product $\imath\mathbf{B}$ with **B** a 2-vector
geom23(B,T).m	Computes the geometric product of bivector **B** with trivector **T**
geom32(T,B).m	Computes the geometric product of trivector **T** with bivector **B**
rspinor(φ,a).m	Computes the spinor with φ as angle and **a** as axis of rotation (1-vector)

APPENDIX C: COMPUTATIONS

The geometrical algebra was implemented in a self-written program which consists of a main m-file (MATLAB file) and eight other auxiliary m-files. These help files are necessary to compute the geometric products (e.g., total geometric and the separate outer product). The main file ($\approx$1 kbyte in size) contains the general computational scheme and uses the functions of the auxiliary files (total of 2.61 kbytes) in its subroutines (Table II).

Calculations are also possible with the MATLAB package GABLE,[25] which was specifically developed for the purpose of computing with geometric algebra. It is available as freeware on the internet. There also exists a variety of other software (freeware and commercial) which is available for performing this type of calculations.

The general strategy for the calculation of a test case was as follows. Firstly, we determined all the geometric parameters $\{l_i, \theta_i, \varphi_i\}$ of the main chain of a small protein (oxymyoglobin). We calculated bond lengths, valence, and dihedral angles by using the data of a crystallographic coordinate file (pdb file in our case). We did not use geometric algebra for those calculations. Afterwards the geometrical parameters were used in the computations which involved geometric algebra. Position coordinates were calculated as pointed out in Sec. III B. The obtained coordinates were aligned (short m-file in MATLAB) with the ones of the pdb file: We superimposed the first three atoms of both chains. We then calculated the root-mean-square deviation between the two sets of coordinates.

It was our aim to show by this example the correctness of the derived formulas. Such a formal proof is only possible if we can compare a calculation with a known structure. Since the pdb file contains in general no geometrical parameters but only the atom coordinates we had to compute those first. To obtain those parameters it is crucial to not use geometric algebra but classic vector algebra. If we then calculate by geometric algebra the position coordinates the correctness of the formulas is proven or disproved.

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